First Challenge

- Launch one **EC2 instance** using **Amazon Linux 2**.

ec2-user@ip-172-31-33-148:~

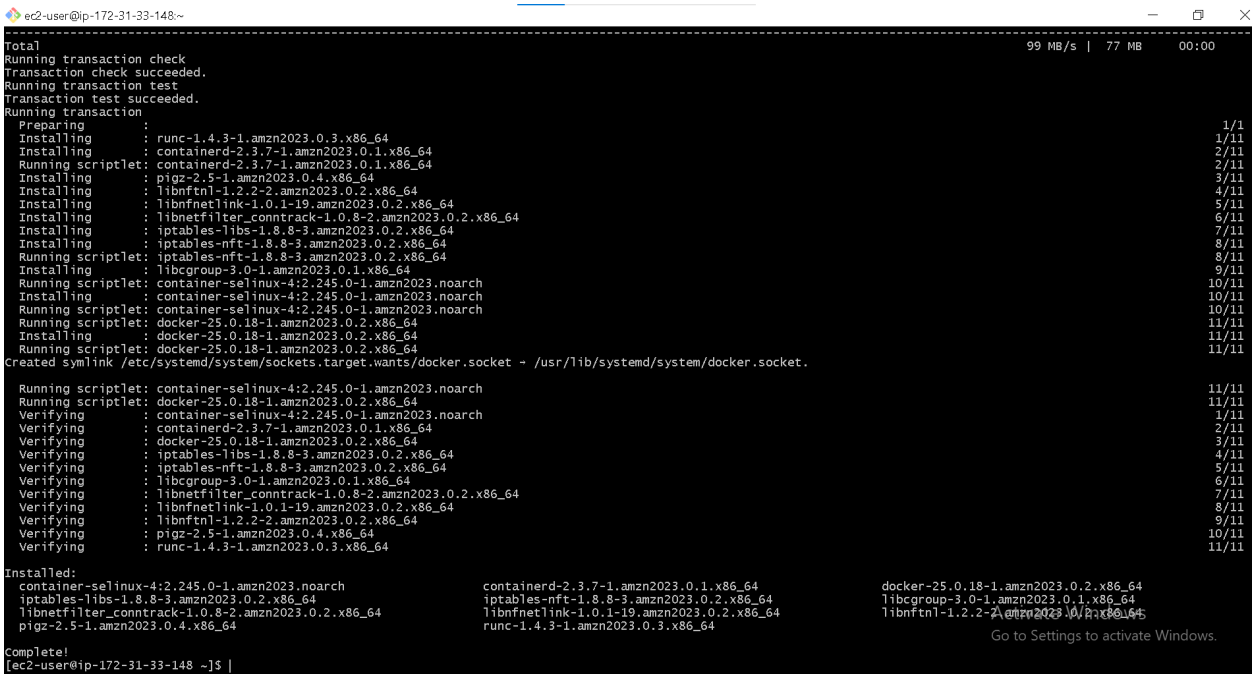
```
Syed Muqeem@DESKTOP-DPRLV22 MINGW64 ~/Downloads
$ ssh -i linux-testing.pem ec2-user@16.178.151.110
The authenticity of host '16.178.151.110 (16.178.151.110)' can't be established.
ED25519 key fingerprint is: SHA256:weUS426UCshGZ50SMp7WD5wl1YaXgsiOdbrD8mqkAP8
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '16.178.151.110' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-33-148 ~]$ |
```

- Install **Docker**.

Step1

Sudo yum install docker

```
[ec2-user@ip-172-31-33-148 ~]$ sudo yum install docker
Amazon Linux 2023 Kernel Livepatch repository                                                
```

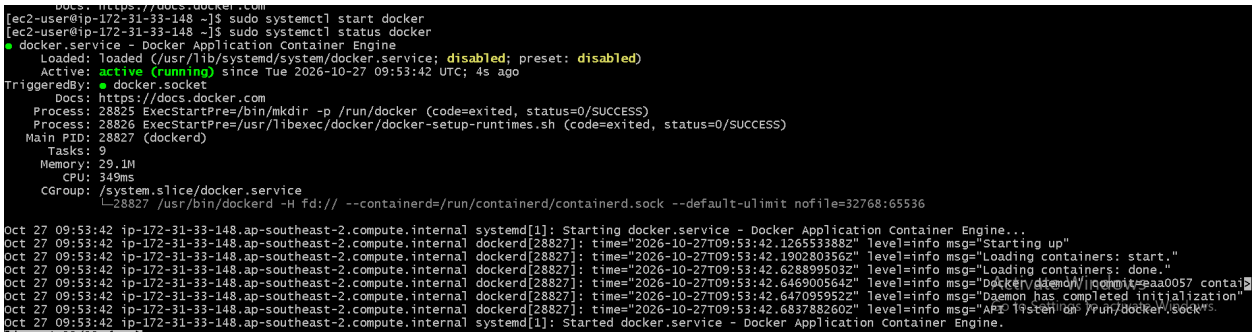


Step2

Start the docker service and check the status

Sudo systemctl start docker

Sudo systemctl status docker



- Install **Jenkins**.

Step1:

Add the Jenkins repo using the following command:

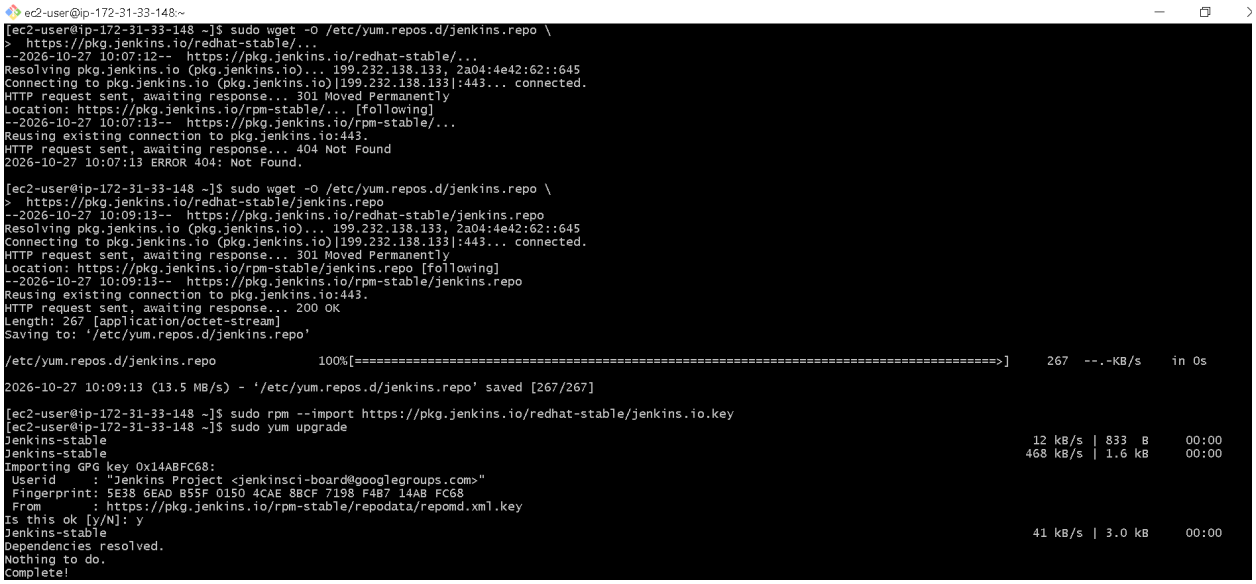
sudo wget -O /etc/yum.repos.d/jenkins.repo \

sudo wget -O /etc/yum.repos.d/jenkins.repo \

Step2: Import a key file from Jenkins-CI to enable installation from the package:

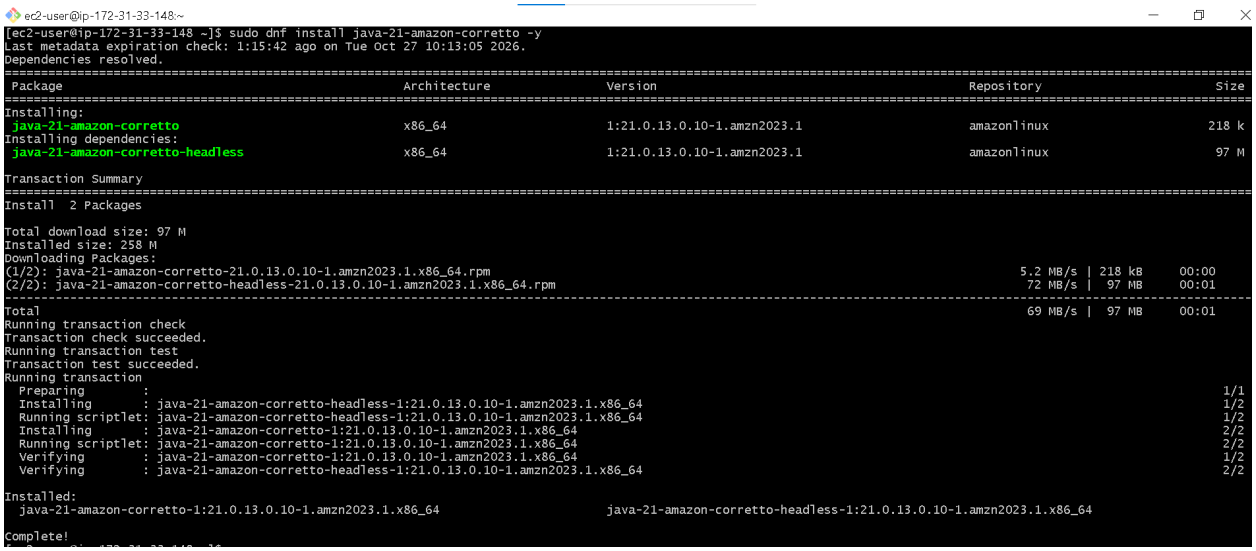
sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io.key

sudo yum upgrade



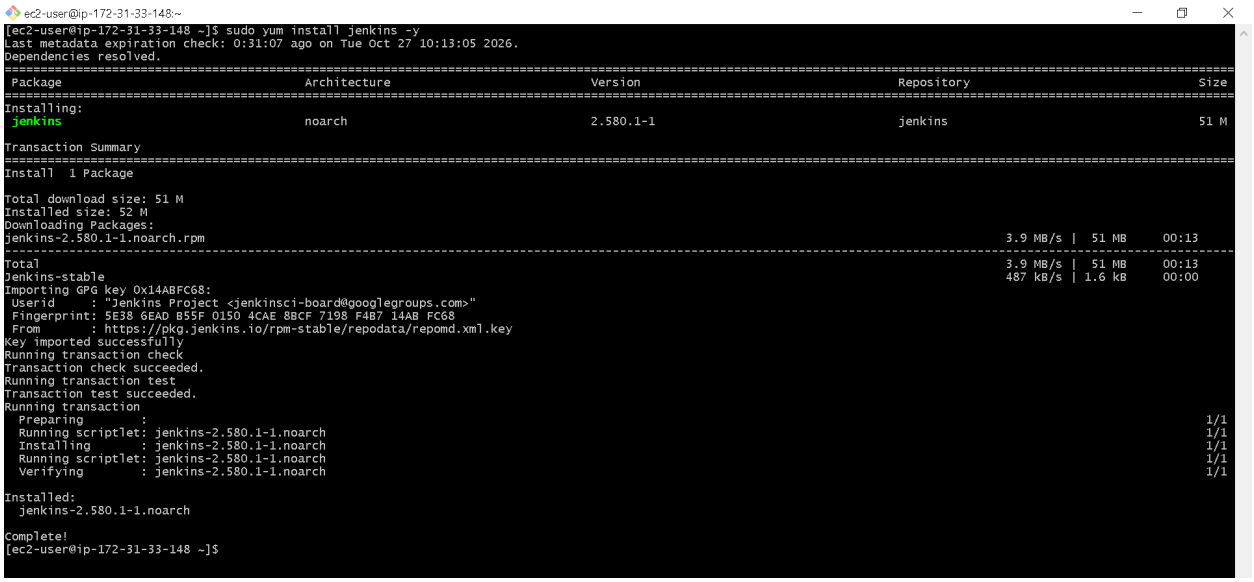
Step3: Install java

sudo dnf install java-21-amazon-corretto -y



Step4: Install jenkins

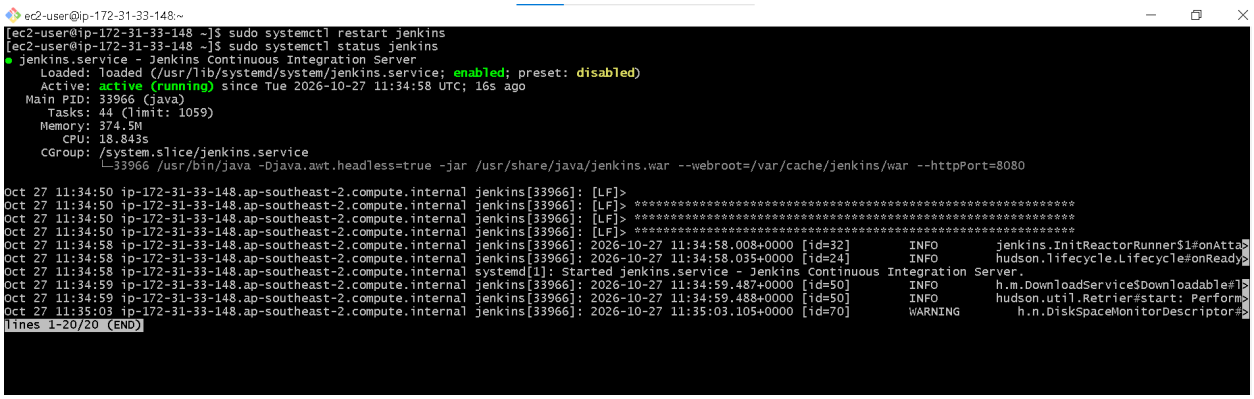
Sudo yum install jenkins -y



Sudo systemctl enable jenkins

Sudo systemctl start jenkins

Sudo systemctl status jenkins



- Install **Apache**.

Step1: to update package index

sudo yum update -y

sudo dnf install httpd -y

```
ec2-user@ip-172-31-33-148:~
[ec2-user@ip-172-31-33-148 ~]$ sudo yum update -y
Last metadata expiration check: 1:28:40 ago on Tue Oct 27 10:13:05 2026.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-33-148 ~]$ sudo yum install httpd -y
Last metadata expiration check: 1:29:10 ago on Tue Oct 27 10:13:05 2026.
Dependencies resolved.
================================================================================
 Package                 Architecture    Version                 Repository    Size
================================================================================
Installing:
 httpd                   x86_64          2.4.69-1.amzn2023.0.2   amazonlinux   45 k
Installing dependencies:
 apr                     x86_64          1.7.5-1.amzn2023.0.4    amazonlinux  129 k
 apr-util                x86_64          1.6.5-1.amzn2023.0.1    amazonlinux  100 k
 apr-util-lmdb           x86_64          1.6.5-1.amzn2023.0.1    amazonlinux   15 k
 generic-logos-httpd     noarch          18.0.0-12.amzn2023.0.3  amazonlinux   19 k
 httpd-core              x86_64          2.4.69-1.amzn2023.0.2   amazonlinux  1.4 M
 httpd-filesystem        noarch          2.4.69-1.amzn2023.0.2   amazonlinux   12 k
 httpd-tools             x86_64          2.4.69-1.amzn2023.0.2   amazonlinux   79 k
 mailcap                 noarch          2.1.49-3.amzn2023.0.3   amazonlinux   33 k
Installing weak dependencies:
 apr-util-openssl        x86_64          1.6.5-1.amzn2023.0.1    amazonlinux   17 k
 mod_http2               x86_64          1:2.4.69-1.amzn2023.0.2 amazonlinux  130 k
 mod_lua                 x86_64          2.4.69-1.amzn2023.0.2   amazonlinux   58 k

Transaction Summary
================================================================================
Install  12 Packages

Total download size: 2.0 M
```

```
Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64
  apr-util-1.6.5-1.amzn2023.0.1.x86_64
  apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64
  apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
  httpd-2.4.69-1.amzn2023.0.2.x86_64
  httpd-core-2.4.69-1.amzn2023.0.2.x86_64
  httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch
  httpd-tools-2.4.69-1.amzn2023.0.2.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch
  mod_http2-1:2.4.69-1.amzn2023.0.2.x86_64
  mod_lua-2.4.69-1.amzn2023.0.2.x86_64

Complete!
```

Activate Windows
Go to Settings to activate Windows.

Step2: start Apache httpd

Sudo systemctl start httpd

sudo systemctl enable httpd

```
ec2-user@ip-172-31-33-148:~
[ec2-user@ip-172-31-33-148 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-33-148 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-33-148 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Tue 2026-10-27 11:44:59 UTC; 31s ago
       Docs: man:httpd.service(8)
   Main PID: 34743 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1059)
     Memory: 14.2M
        CPU: 104ms
     CGroup: /system.slice/httpd.service
             ├─34743 /usr/sbin/httpd -DFOREGROUND
             ├─34771 /usr/sbin/httpd -DFOREGROUND
             ├─34772 /usr/sbin/httpd -DFOREGROUND
             ├─34773 /usr/sbin/httpd -DFOREGROUND
             └─34775 /usr/sbin/httpd -DFOREGROUND

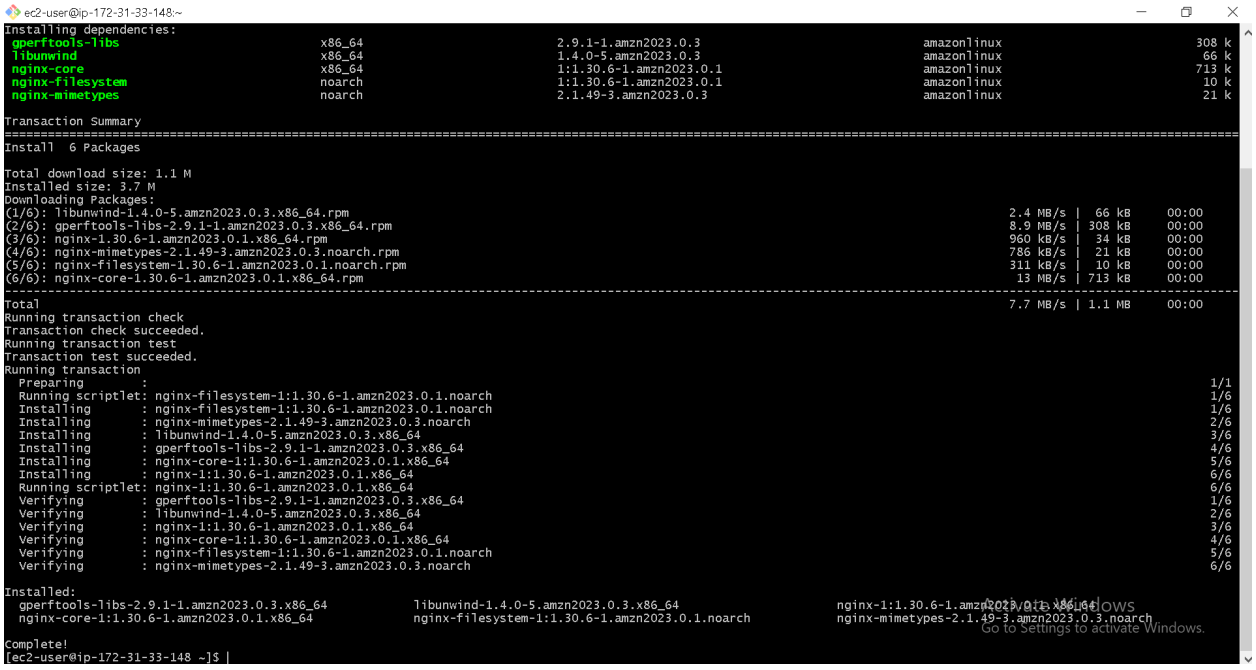
Oct 27 11:44:59 ip-172-31-33-148.ap-southeast-2.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Oct 27 11:44:59 ip-172-31-33-148.ap-southeast-2.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Oct 27 11:44:59 ip-172-31-33-148.ap-southeast-2.compute.internal httpd[34743]: Server configured, listening on: port 80
[ec2-user@ip-172-31-33-148 ~]$
```

- Install **Nginx**.

Step1:Update your package

sudo yum update -y

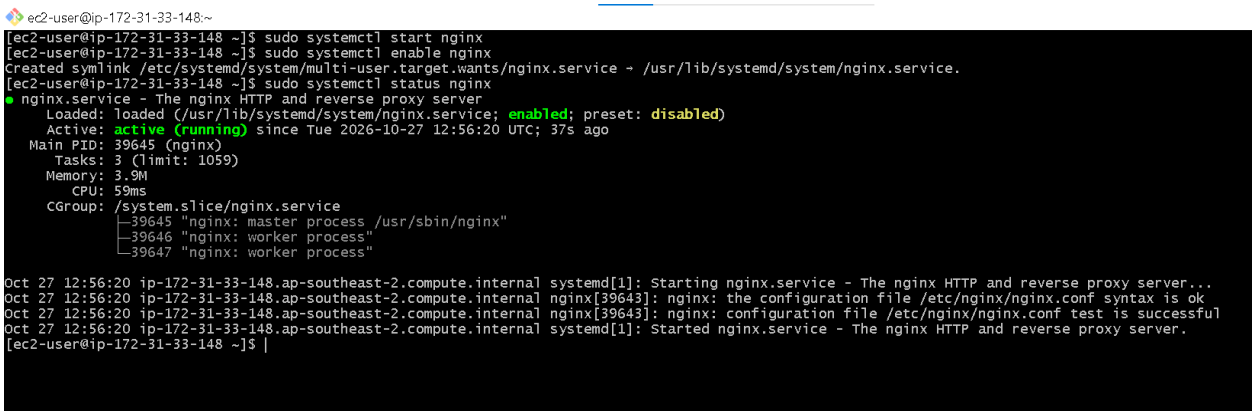
sudo yum install nginx -y



sudo systemctl start nginx

sudo systemctl enable nginx

sudo systemctl status nginx





Welcome to nginx!

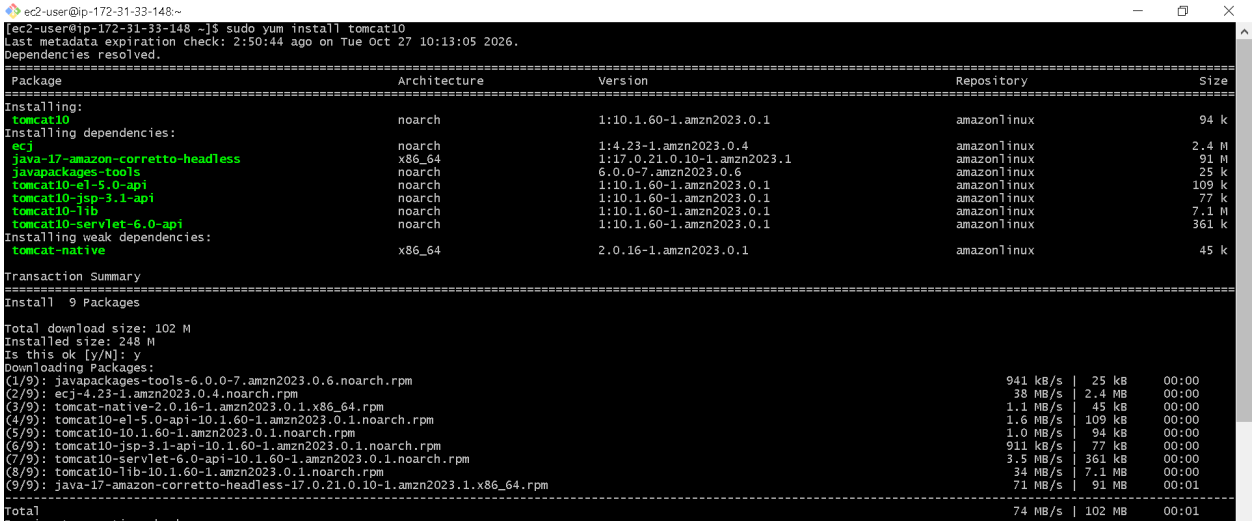
If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

- Install **Apache Tomcat**.

Step1: sudo yum install tomcat10





Step2: after configuring port

Sudo systemctl restart tomcat10

Sudo systemctl status tomcat10

